

Adaptive Selection of the Gaussian Sampling Scale for Monte Carlo Estimation of Loss-Landscape Convergence

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Abstract

Understanding the local behavior of the loss function near a minimum is important both for analyzing neural loss landscapes and for studying how the geometry of the optimization surface changes with the training set size. Recent Monte Carlo approaches to loss-landscape convergence rely on Gaussian sampling around a local minimum, but their performance is highly sensitive to the choice of the sampling scale. In particular, if the sampling variance is too large, sampled points leave the neighborhood where the loss is well approximated by its second-order Taylor expansion; if it is too small, the sampling becomes overly conservative. In this work, we propose an adaptive rule for selecting the Gaussian sampling scale based on local curvature information. Using the Hessian at a minimum and a local regularity condition on the loss landscape, we derive an explicit estimate of the radius of the local quadratic regime and obtain a closed-form expression for the largest admissible sampling scale that stays inside this region with prescribed probability. We further validate the proposed rule experimentally on image classification with ResNet-18 on CIFAR-10. The results show that the selected scale is well aligned with the region where the quadratic approximation remains accurate and that the admissible scale decreases as the dimension of the dominant-curvature subspace increases. Overall, the proposed method provides a practical geometry-aware calibration of Gaussian sampling for local loss-landscape analysis and for Monte Carlo estimation of loss-landscape convergence.

Keywords: loss landscape, Hessian, Gaussian sampling, Taylor approximation, Monte Carlo estimation

1 Introduction

In statistical learning theory, the behavior of a model is fundamentally linked to the size of the training dataset [1]. For parametric models, this raises the question of how large a dataset must be before the local geometry of the loss landscape stabilizes. One way to formalize this question is to study the convergence of the loss landscape as the number of

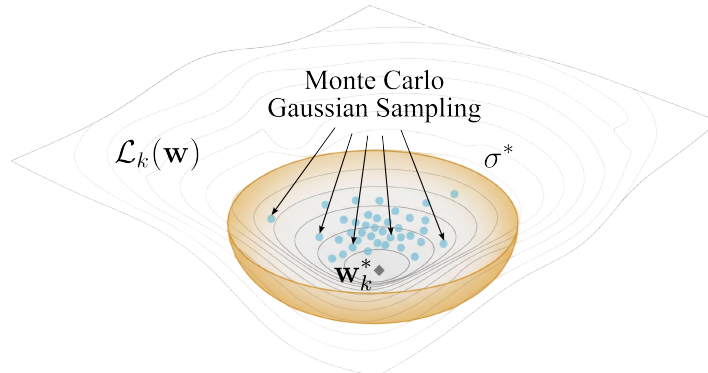


Figure 1: Sampling-scale selection principle: Gaussian perturbations should remain inside a characteristic neighborhood around the minimum $\mathcal{L}(\mathbf{w}^*)$ where the second-order Taylor approximation is sufficiently accurate.

training examples increases [2]. Informally, once the geometry of the loss surface near a minimum changes only weakly when new samples are added, the current dataset size may be viewed as sufficient for stable training.

Recent work proposed quantitative criteria for measuring such convergence and suggested estimating them by Gaussian sampling around a local minimum of the loss function [3, 4]. More recently, Quadratic Monte-Carlo and Gaussian-moment quadratic estimators were introduced as computationally more efficient alternatives based on second-order local approximations. These estimators are attractive because they replace repeated evaluation of the full loss landscape by quantities derived from its local quadratic structure.

However, their practical use requires more than asymptotic convergence guarantees. In order for quadratic estimators to be reliable, the sampled perturbations must remain inside a neighborhood where the loss is well approximated by its second-order Taylor expansion around the minimum [5]. If the sampling scale is chosen too large, a substantial fraction of sampled points leaves this local quadratic regime, and the resulting estimates no longer accurately reflect the underlying geometry of the loss landscape. If the scale is chosen too small, the sampling becomes overly conservative and fails to probe the neighborhood in a meaningful way.

This motivates the central question of the present work: how should one choose the Gaussian sampling scale so that sampled points remain inside the local quadratic regime with high probability? We address this question by linking the admissible sampling scale to local geometric properties of the loss landscape near a minimum. In particular, we derive a characteristic radius of validity for the second-order Taylor approximation and use it to obtain an adaptive rule for selecting the Gaussian sampling scale in the dominant-curvature subspace of the Hessian. The resulting construction provides a principled calibration of Gaussian sampling for local loss-landscape analysis.

Our contributions are as follows:

- We derive an explicit estimate of the radius R_ψ of a neighborhood around a local

minimum in which the second-order Taylor approximation remains accurate up to a prescribed tolerance.

- We obtain a closed-form expression for the largest Gaussian sampling scale σ^* that keeps perturbations inside this quadratic-validity region with probability at least $1 - \varepsilon$.
- We validate the proposed scale-selection rule empirically on ResNet-18 trained on CIFAR-10 and show that it identifies a regime in which the quadratic approximation remains accurate, while larger perturbations quickly leave this regime.

2 Related Work

Loss-landscape convergence under increasing sample size. Understanding how the behavior of a parametric model depends on the dataset size is a classical problem in statistical learning theory [1]. Recent works studied the convergence of the loss landscape as the number of training examples increases and introduced quantitative criteria for measuring this effect, including the absolute criterion Δ_1 and the quadratic criterion Δ_2 [2, 3, 6]. Later, Gaussian Monte Carlo sampling around a local minimum was proposed as a practical way to estimate such criteria numerically [4]. More recently, Quadratic Monte-Carlo and Gaussian-moment quadratic estimators were introduced as more computationally efficient alternatives. However, their practical validity relies on sampling within a region where the second-order Taylor approximation is sufficiently accurate. Our work addresses this issue directly by deriving an adaptive rule for choosing the Gaussian sampling scale so that the sampled points remain in the local quadratic regime with high probability.

Local geometry of neural loss landscapes. A large body of work has analyzed the local geometry of neural loss landscapes, both theoretically and empirically, including visualization methods, spectral analysis, and curvature-based characterizations of minima [7, 8, 9, 10, 11, 12]. These studies consistently show that neural loss landscapes are highly anisotropic: a small number of directions exhibit significant curvature, while much of the Hessian spectrum remains close to zero [13, 14, 15]. This observation motivates the use of dominant-curvature subspaces spanned by the leading Hessian eigenvectors [4]. Our method builds on this idea. Instead of using such subspaces only for visualization or probing, we derive an explicit geometry-aware scale-selection rule for Gaussian sampling within them.

Curvature, minima geometry, and optimization. The geometry of minima has also been studied in connection with optimization and generalization. In particular, the distinction between flat and sharp minima has played a central role in understanding the behavior of deep networks and stochastic optimization methods [16, 17, 18, 19]. Several works analyzed how SGD and related methods interact with the curvature structure of the

loss landscape and how optimization trajectories behave in the full parameter space and in high-curvature subspaces [20, 21, 22, 23, 24]. These results highlight the importance of understanding local geometry near minima and of analyzing loss variation along structured directions. In this context, our contribution is complementary: we provide an explicit estimate of the radius of validity of the second-order approximation and use it to calibrate Gaussian perturbations for local loss-landscape analysis.

3 Problem Statement

Let $\mathcal{L}_k : \mathbb{R}^N \rightarrow \mathbb{R}$ denote the loss function of a parametric model $f_{\mathbf{w}} : \mathbb{X} \rightarrow \mathbb{Y}$ trained on a dataset $\mathfrak{D}_k = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^k$ consisting of k i.i.d. samples from data distribution $p(\mathbf{x}, \mathbf{y})$, and let

$$\mathbf{w}_k^* \in \arg \min_{\mathbf{w} \in \mathbb{R}^N} \mathcal{L}_k(\mathbf{w})$$

be a local minimum of $\mathcal{L}_k$.

Our goal is to study the local geometry of the loss landscape near $\mathbf{w}_k^*$ and to use this geometry for constructing practical Monte Carlo estimators of loss-landscape convergence as the dataset size increases. In particular, we are interested in sampling perturbations around $\mathbf{w}_k^*$ in a way that remains informative while preserving the validity of the second-order Taylor approximation.

Assume that $\mathcal{L}_k$ is twice continuously differentiable in a neighborhood of $\mathbf{w}_k^*$. Then for any sufficiently small perturbation $\mathbf{h} \in \mathbb{R}^N$, we have the second-order Taylor expansion

$$\mathcal{L}_k(\mathbf{w}_k^* + \mathbf{h}) = \mathcal{L}_k(\mathbf{w}_k^*) + \nabla \mathcal{L}_k(\mathbf{w}_k^*)^\top \mathbf{h} + \frac{1}{2} \mathbf{h}^\top \mathbf{H}_k \mathbf{h} + \rho_k(\mathbf{h}),$$

where $\mathbf{H}_k = \nabla^2 \mathcal{L}_k(\mathbf{w}_k^*)$ is the Hessian at the minimum, and $\rho_k(\mathbf{h})$ is the remainder term. Since $\mathbf{w}_k^*$ is a local minimum, we have $\nabla \mathcal{L}_k(\mathbf{w}_k^*) = \mathbf{0}$, so the local behavior of the loss is governed by

$$\mathcal{L}_k(\mathbf{w}_k^* + \mathbf{h}) = \mathcal{L}_k(\mathbf{w}_k^*) + \frac{1}{2} \mathbf{h}^\top \mathbf{H}_k \mathbf{h} + \rho_k(\mathbf{h}).$$

In high-dimensional models, the local geometry of the loss landscape is typically highly anisotropic. A substantial fraction of directions has very small curvature, while the dominant variation is concentrated in a relatively low-dimensional subspace associated with the leading eigenvalues of the Hessian. For this reason, instead of sampling perturbations in the full parameter space $\mathbb{R}^N$, we consider sampling in the dominant-curvature subspace.

Let $\lambda_1^{(k)} \geq \lambda_2^{(k)} \geq \dots \geq \lambda_N^{(k)}$ be the eigenvalues of $\mathbf{H}_k$, and let

$$\mathbf{U}_{k,d} = [\mathbf{u}_1^{(k)}, \dots, \mathbf{u}_d^{(k)}] \in \mathbb{R}^{N \times d}$$

be the matrix whose columns are the eigenvectors corresponding to the d largest eigenvalues. We define the dominant-curvature subspace by

$$\mathcal{S}_{k,d} = \text{span}\{\mathbf{u}_1^{(k)}, \dots, \mathbf{u}_d^{(k)}\}.$$

We then consider Gaussian sampling in this subspace:

$$\mathbf{w} = \mathbf{w}_k^* + \mathbf{U}_{k,d}\mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_d),$$

where $\sigma > 0$ is the sampling scale. Equivalently, the perturbation is $\mathbf{h} = \mathbf{U}_{k,d}\mathbf{z} \in \mathcal{S}_{k,d}$.

The key difficulty is to choose σ appropriately. If σ is too small, the sampled points remain extremely close to the minimum and do not probe the local landscape in a meaningful way. If σ is too large, a non-negligible fraction of the samples leaves the region where the second-order Taylor approximation is accurate, which makes quadratic estimators unreliable.

To formalize this trade-off, let $\psi > 0$ denote the desired relative accuracy of the quadratic approximation, and suppose that there exists a radius $R_{k,\psi} > 0$ such that for all perturbations $\mathbf{h} \in \mathcal{S}_{k,d}$ satisfying $\|\mathbf{h}\|_2 \leq R_{k,\psi}$, the remainder term is controlled by the quadratic term:

$$|\rho_k(\mathbf{h})| \leq \psi \cdot \frac{1}{2} \mathbf{h}^\top \mathbf{H}_k \mathbf{h}.$$

Thus, the ball of radius $R_{k,\psi}$ centered at $\mathbf{w}_k^*$ represents the region in which the second-order approximation is sufficiently accurate for our purposes.

We would like the Gaussian sampling distribution to remain inside this region with high probability. Given a tolerance level $\varepsilon \in (0, 1)$, we therefore seek values of σ such that

$$\mathbb{P}(\|\mathbf{w} - \mathbf{w}_k^*\|_2 \leq R_{k,\psi}) \geq 1 - \varepsilon.$$

This leads to the following problem: determine the maximal admissible sampling scale

$$\sigma_{k,d}^* = \sup \{ \sigma > 0 : \mathbb{P}(\|\mathbf{w} - \mathbf{w}_k^*\|_2 \leq R_{k,\psi}) \geq 1 - \varepsilon \}.$$

In other words, we seek an explicit expression, or a practically computable estimate, for $\sigma_{k,d}^*$ in terms of local geometric characteristics of the loss landscape near $\mathbf{w}_k^*$.

The remainder of the paper addresses this problem in two steps. First, we derive a radius of validity for the quadratic approximation in the dominant-curvature subspace under a local regularity assumption on the Hessian. Second, we use this radius to obtain a closed-form expression for the largest Gaussian sampling scale that keeps the sampled points inside the quadratic regime with probability at least $1 - \varepsilon$.

4 Method

We derive an adaptive rule for selecting the Gaussian sampling scale in the dominant-curvature subspace near a local minimum. The construction consists of three steps. First, we characterize a neighborhood around the minimum in which the second-order Taylor approximation is accurate up to a prescribed tolerance. Second, we convert this neighborhood radius into a maximal admissible Gaussian sampling scale. Third, we describe the practical estimator used in experiments.

4.1 Setup and assumptions

For simplicity of notation, in this section we omit the dataset-size index k and write $\mathcal{L} = \mathcal{L}_k$, $\mathbf{w}^* = \mathbf{w}_k^*$, $\mathbf{H} = \mathbf{H}_k$, $\mathbf{U}_d = \mathbf{U}_{k,d}$. Let $\mathbf{H}_d = \mathbf{U}_d^\top \mathbf{H} \mathbf{U}_d \in \mathbb{R}^{d \times d}$ denote the restriction of the Hessian to the dominant-curvature subspace $\mathcal{S}_d = \text{span}(\mathbf{U}_d)$. Since $\mathbf{w}^*$ is a local minimum, we have $\nabla \mathcal{L}(\mathbf{w}^*) = \mathbf{0}$. Therefore, for every sufficiently small perturbation $\mathbf{h} \in \mathbb{R}^N$, the second-order Taylor expansion takes the form

$$\mathcal{L}(\mathbf{w}^* + \mathbf{h}) = \mathcal{L}(\mathbf{w}^*) + \frac{1}{2} \mathbf{h}^\top \mathbf{H} \mathbf{h} + \rho(\mathbf{h}),$$

where $\rho(\mathbf{h})$ is the Taylor remainder. We assume local regularity of the Hessian around the minimum.

Assumption 1. There exists $r > 0$ such that the Hessian $\nabla^2 \mathcal{L}$ is Lipschitz continuous in the ball $\mathcal{B}_r(\mathbf{w}^*)$ with constant L_H , i.e.,

$$\|\nabla^2 \mathcal{L}(\mathbf{u}) - \nabla^2 \mathcal{L}(\mathbf{v})\|_2 \leq L_H \|\mathbf{u} - \mathbf{v}\|_2 \quad \forall \mathbf{u}, \mathbf{v} \in \mathcal{B}_r(\mathbf{w}^*).$$

Assumption 2. The restricted Hessian $\mathbf{H}_d$ is positive definite: $\lambda_{\min}(\mathbf{H}_d) > 0$.

We consider perturbations restricted to the dominant-curvature subspace: $\mathbf{h} = \mathbf{U}_d \mathbf{z}$, $\mathbf{z} \in \mathbb{R}^d$. Because the columns of $\mathbf{U}_d$ are orthonormal, $\|\mathbf{h}\|_2 = \|\mathbf{z}\|_2$.

4.2 Radius of the local quadratic regime

We first derive a radius within which the second-order Taylor approximation is accurate relative to the quadratic term.

Proposition 3. Let $\psi > 0$ be a prescribed tolerance level. Under Assumptions 1–2, for every $\mathbf{z} \in \mathbb{R}^d$ satisfying

$$\|\mathbf{z}\|_2 \leq R_\psi, \quad R_\psi = \min \left\{ r, \frac{3\psi \lambda_{\min}(\mathbf{H}_d)}{L_H} \right\},$$

the perturbation $\mathbf{h} = \mathbf{U}_d \mathbf{z}$ satisfies

$$|\rho(\mathbf{U}_d \mathbf{z})| \leq \psi \cdot \frac{1}{2} \mathbf{z}^\top \mathbf{H}_d \mathbf{z}.$$

Equivalently,

$$|\rho(\mathbf{h})| \leq \psi \cdot \frac{1}{2} \mathbf{h}^\top \mathbf{H} \mathbf{h} \quad \text{for all } \mathbf{h} \in \mathcal{S}_d \text{ with } \|\mathbf{h}\|_2 \leq R_\psi.$$

Proof. Using the integral form of the Taylor remainder, we have

$$\rho(\mathbf{h}) = \int_0^1 (1-t) \mathbf{h}^\top (\nabla^2 \mathcal{L}(\mathbf{w}^* + t\mathbf{h}) - \nabla^2 \mathcal{L}(\mathbf{w}^*)) \mathbf{h} dt.$$

Let $\|\mathbf{h}\|_2 \leq r$. Then for every $t \in [0, 1]$,

$$\|\mathbf{w}^* + t\mathbf{h} - \mathbf{w}^*\|_2 = t\|\mathbf{h}\|_2 \leq r,$$

so $\mathbf{w}^* + t\mathbf{h} \in \mathcal{B}_r(\mathbf{w}^*)$, and Assumption 1 applies. Therefore,

$$\|\nabla^2 \mathcal{L}(\mathbf{w}^* + t\mathbf{h}) - \nabla^2 \mathcal{L}(\mathbf{w}^*)\|_2 \leq L_H t \|\mathbf{h}\|_2.$$

Substituting this into the integral representation yields

$$|\rho(\mathbf{h})| \leq \int_0^1 (1-t) \|\mathbf{h}\|_2^2 \|\nabla^2 \mathcal{L}(\mathbf{w}^* + t\mathbf{h}) - \nabla^2 \mathcal{L}(\mathbf{w}^*)\|_2 dt \leq L_H \|\mathbf{h}\|_2^3 \int_0^1 t(1-t) dt = \frac{L_H}{6} \|\mathbf{h}\|_2^3.$$

Now let $\mathbf{h} = \mathbf{U}_d \mathbf{z}$. Then

$$\mathbf{h}^\top \mathbf{H} \mathbf{h} = \mathbf{z}^\top \mathbf{U}_d^\top \mathbf{H} \mathbf{U}_d \mathbf{z} = \mathbf{z}^\top \mathbf{H}_d \mathbf{z} \geq \lambda_{\min}(\mathbf{H}_d) \|\mathbf{z}\|_2^2.$$

Since $\|\mathbf{h}\|_2 = \|\mathbf{z}\|_2$, we obtain

$$\mathbf{h}^\top \mathbf{H} \mathbf{h} \geq \lambda_{\min}(\mathbf{H}_d) \|\mathbf{h}\|_2^2.$$

Hence the inequality

$$\frac{L_H}{6} \|\mathbf{h}\|_2^3 \leq \psi \cdot \frac{1}{2} \lambda_{\min}(\mathbf{H}_d) \|\mathbf{h}\|_2^2$$

is sufficient for

$$|\rho(\mathbf{h})| \leq \psi \cdot \frac{1}{2} \mathbf{h}^\top \mathbf{H} \mathbf{h}.$$

For $\mathbf{h} \neq \mathbf{0}$, this reduces to

$$\|\mathbf{h}\|_2 \leq \frac{3\psi \lambda_{\min}(\mathbf{H}_d)}{L_H}.$$

Combining this with the requirement $\|\mathbf{h}\|_2 \leq r$ completes the proof. $\square$

Proposition 3 identifies a radius R_ψ such that perturbations inside $\mathcal{B}_{R_\psi}(\mathbf{w}^*) \cap \mathcal{S}_d$ belong to the local quadratic regime with relative tolerance ψ .

4.3 Admissible Gaussian sampling scale

We now convert the radius R_ψ into a rule for selecting the Gaussian sampling scale. Recall that perturbations are sampled according to $\mathbf{w} = \mathbf{w}^* + \mathbf{U}_d \mathbf{z}$, $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_d)$. Since $\mathbf{U}_d$ has orthonormal columns, $\|\mathbf{w} - \mathbf{w}^*\|_2^2 = \|\mathbf{U}_d \mathbf{z}\|_2^2 = \|\mathbf{z}\|_2^2$. Moreover, $\|\mathbf{z}\|_2^2 / \sigma^2 \sim \chi^2(d)$. The following theorem gives the largest Gaussian sampling scale such that perturbations remain inside the local quadratic regime with probability at least $1 - \varepsilon$.

Theorem 4. Let $\varepsilon \in (0, 1)$, and let R_ψ be the quadratic-validity radius from Proposition 3. Then the maximal admissible Gaussian sampling scale

$$\sigma^* = \sup \{ \sigma > 0 : \mathbb{P}(\|\mathbf{w} - \mathbf{w}^*\|_2 \leq R_\psi) \geq 1 - \varepsilon \}$$

is given by

$$\sigma^* = \frac{R_\psi}{\sqrt{\chi_{1-\varepsilon}^2(d)}}.$$

Proof. Since $\mathbf{w} - \mathbf{w}^* = \mathbf{U}_d \mathbf{z}$ and $\mathbf{U}_d$ has orthonormal columns, we have

$$\|\mathbf{w} - \mathbf{w}^*\|_2^2 = \|\mathbf{z}\|_2^2.$$

Therefore,

$$\mathbb{P}(\|\mathbf{w} - \mathbf{w}^*\|_2 \leq R_\psi) = \mathbb{P}(\|\mathbf{z}\|_2^2 \leq R_\psi^2).$$

Since $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_d)$, it follows that

$$\frac{\|\mathbf{z}\|_2^2}{\sigma^2} \sim \chi^2(d).$$

Hence,

$$\mathbb{P}(\|\mathbf{z}\|_2^2 \leq R_\psi^2) = \mathbb{P}\left(\chi^2(d) \leq \frac{R_\psi^2}{\sigma^2}\right).$$

The largest σ such that this probability is at least $1 - \varepsilon$ is attained when

$$\frac{R_\psi^2}{\sigma^2} = \chi_{1-\varepsilon}^2(d),$$

that is,

$$\sigma^* = \frac{R_\psi}{\sqrt{\chi_{1-\varepsilon}^2(d)}}.$$

Substituting the expression for R_ψ from Proposition 3 completes the proof. $\square$

Theorem 4 makes the trade-off explicit. The admissible sampling scale decreases when the tolerance ψ becomes stricter, when the weakest curvature in the selected subspace decreases, or when the Hessian varies more rapidly around the minimum. It also decreases with the subspace dimension d through the chi-square quantile.

4.4 Practical estimation

The theoretical expression above depends on local quantities that are not directly available in large-scale models, in particular the Hessian Lipschitz constant L_H . We therefore use a practical approximation based on finite differences of the restricted Hessian along the dominant-curvature subspace.

For $\boldsymbol{\xi} \in \mathbb{R}^d$, define $\mathbf{H}_d(\boldsymbol{\xi}) = \mathbf{U}_d^\top \nabla^2 \mathcal{L}(\mathbf{w}^* + \mathbf{U}_d \boldsymbol{\xi}) \mathbf{U}_d$. Thus, $\mathbf{H}_d(\mathbf{0}) = \mathbf{H}_d$. For a small scalar $t > 0$ and a unit vector $\mathbf{v} \in \mathbb{R}^d$, define the directional estimator

$$\widehat{L}_H(\mathbf{v}; t) = \frac{\|\mathbf{H}_d(t\mathbf{v}) - \mathbf{H}_d(\mathbf{0})\|_2}{t}.$$

In practice, we sample M random unit directions $\mathbf{v}_1, \dots, \mathbf{v}_M$ in $\mathbb{R}^d$ and set

$$\widehat{L}_H = \max_{1 \leq m \leq M} \widehat{L}_H(\mathbf{v}_m; t).$$

Replacing L_H by $\widehat{L}_H$ yields the practical estimator

$$\widehat{\sigma}^* = \frac{1}{\sqrt{\chi_{1-\varepsilon}^2(d)}} \min \left\{ r, \frac{3\psi \lambda_{\min}(\mathbf{H}_d)}{\widehat{L}_H} \right\}.$$

This estimator is the quantity used in our experiments. It provides an adaptive, geometry-aware choice of the Gaussian sampling scale calibrated to the local curvature structure of the loss landscape near the minimum.

5 Experiments

We evaluate the proposed adaptive sampling rule on a ResNet-18 model trained on CIFAR-10. Our experimental study has two goals. First, we investigate how the admissible sampling scale σ^* depends on the dimension d of the dominant-curvature subspace. Second, we verify that the resulting scale indeed keeps Gaussian perturbations inside a region where the second-order Taylor approximation remains accurate.

5.1 Experimental setup

After training ResNet-18 on CIFAR-10, we consider the obtained solution $\mathbf{w}^*$ as the reference minimum. At this point, we compute the leading eigenvalues and eigenvectors of the Hessian using power iteration combined with Hessian-vector products. For each chosen subspace dimension d , we construct the dominant-curvature subspace spanned by the top- d eigenvectors, $\mathcal{S}_d = \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_d\}$, and the corresponding restricted Hessian $\mathbf{H}_d = \mathbf{U}_d^\top \mathbf{H} \mathbf{U}_d$.

Using the quantities defined in Section 4, we compute the adaptive sampling scale σ^* for each value of d . We then study the behavior of σ^* as a function of the subspace dimension and analyze the quality of the resulting quadratic approximation under Gaussian sampling in $\mathcal{S}_d$.

5.2 Dependence of the admissible scale on the subspace dimension

We first examine how the estimated admissible sampling scale varies with the dimension of the dominant-curvature subspace. For each tested value of d , we compute σ^* and report the normalized quantity $\sigma^*(d)/\sigma^*(10)$ for ease of comparison.

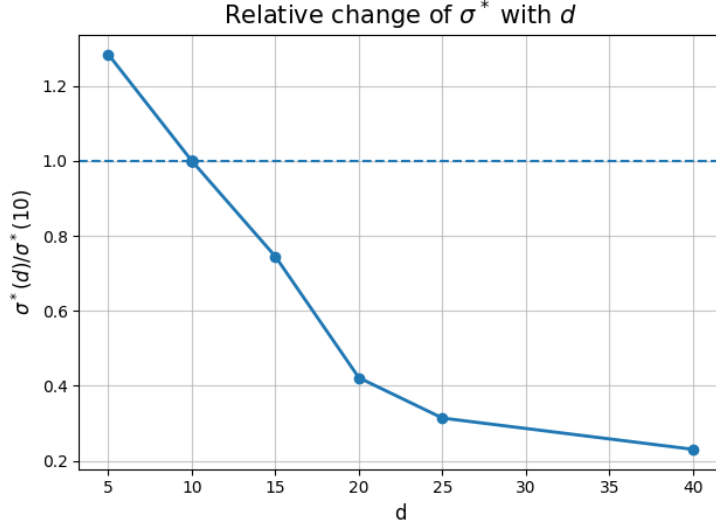


Figure 2: Normalized admissible sampling scale $\sigma^*(d)/\sigma^*(10)$ as a function of the dominant-curvature subspace dimension d .

Figure 2 shows that the admissible sampling scale decreases rapidly as d increases. This behavior is consistent with the theoretical expression derived in the previous section. As the dimension grows, the chi-square quantile increases, while the weakest curvature direction inside the selected subspace typically becomes smaller. Both effects reduce the largest scale that still keeps Gaussian samples inside the local quadratic regime with high probability.

Empirically, this suggests that relatively small values of d , such as $d \approx 10$, provide a favorable operating point: they capture the dominant curvature directions while still allowing a nontrivial sampling radius.

5.3 Validation of the quadratic regime

We next test whether the estimated scale σ^* indeed identifies a regime in which the second-order Taylor approximation is accurate. To this end, we sample perturbations in the dominant-curvature subspace according to $\mathbf{w} = \mathbf{w}^* + \mathbf{U}_d \mathbf{z}$, $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_d)$, with $\sigma = \alpha \sigma^*$, where $\alpha > 0$ is a multiplicative scaling factor. In the experiments, we consider $\alpha \in [1, 101]$ with step size 10.

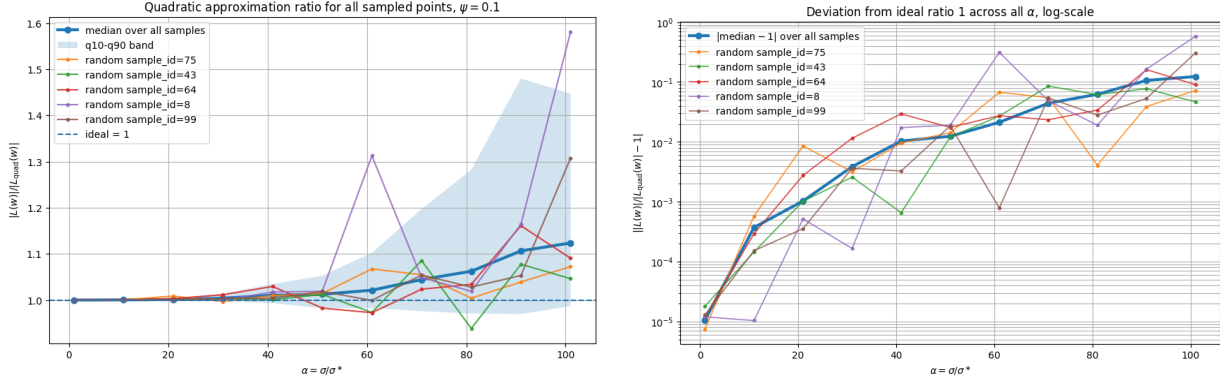
To quantify the quality of the quadratic approximation, for each sampled point $\mathbf{w}$ we compute

$$Q(\mathbf{w}) = \frac{|\mathcal{L}(\mathbf{w})|}{|\mathcal{L}(\mathbf{w}^*) + \nabla \mathcal{L}(\mathbf{w}^*)^\top \mathbf{U}_d \mathbf{z} + \frac{1}{2} \mathbf{z}^\top \mathbf{H}_d \mathbf{z}| + \delta},$$

where $\delta > 0$ is a small constant added for numerical stability. The quantity $Q(\mathbf{w})$ measures the ratio between the true loss value at $\mathbf{w}$ and its second-order Taylor approximation around $\mathbf{w}^*$.

If the sampling scale is well calibrated, then $Q(\mathbf{w})$ should remain close to 1. As the scale increases, the sampled points move farther away from the minimum, and the approximation error is expected to grow.

We summarize the results by plotting the median value of $Q(\mathbf{w})$ as a function of the scaling factor α , and additionally report $|Q(\mathbf{w}) - 1|$ to visualize the deviation from the quadratic regime more clearly.



(a) Median value of $Q(\mathbf{w})$ as a function of α . (b) Deviation $|Q(\mathbf{w}) - 1|$ as a function of α in log scale.

Figure 3: Validation of the quadratic regime under scaled Gaussian perturbations $\sigma = \alpha\sigma^*$.

Figure 3 shows that when α is close to 1, the ratio $Q(\mathbf{w})$ remains close to 1, indicating that the sampled points lie in a region where the quadratic approximation is accurate. As α increases, the deviation from 1 grows rapidly, showing that the perturbations progressively leave the local quadratic regime.

These results provide empirical support for the proposed adaptive scale selection rule. In particular, they show that the theoretically motivated scale σ^* identifies a meaningful boundary between a stable quadratic regime and a regime where second-order approximations become unreliable. At the same time, the plots suggest that the estimate may be somewhat conservative in practice, which is reasonable given that the local regularity quantities entering the formula are themselves estimated numerically.

6 Discussion

The proposed method links the admissible Gaussian sampling scale to local geometric properties of the loss landscape near a minimum. In particular, the scale σ^* is controlled by the weakest curvature in the selected dominant-curvature subspace and by the local variation of the Hessian. This makes the choice of the sampling scale adaptive to the local structure of the landscape rather than fixed a priori.

The analysis also clarifies the role of the subspace dimension d . As d increases, the admissible scale typically decreases because the chi-square quantile grows and the smallest

eigenvalue of the restricted Hessian tends to become smaller. This explains why relatively low-dimensional dominant-curvature subspaces may be preferable in practice.

The experiments are consistent with this picture. They show both that σ^* decreases with d and that the estimated scale identifies a regime in which the quadratic approximation remains accurate. At the same time, the current study is local and limited to a single architecture–dataset pair, so broader validation remains an important direction for future work.

7 Conclusion

We proposed an adaptive rule for choosing the Gaussian sampling scale near a local minimum of the loss function. The method is based on an estimate of the radius of the local quadratic regime and leads to an explicit expression for the largest admissible sampling scale with prescribed confidence.

Experiments on ResNet-18 trained on CIFAR-10 support the proposed approach. In particular, the results show that the selected scale is aligned with the region where the second-order Taylor approximation remains accurate, while larger perturbations quickly leave this regime. Overall, the method provides a practical geometry-aware calibration of Gaussian sampling for studying local loss-landscape structure.

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